Difference Array Concepts & Qns

Video - 8

Motivation :-

Protect your focus like it is your most valuable asset, because in a world full of noise, deep work is a superpower.

MIK...

Difference Array with Jumps

5 3 7 6 4 3

0 1 2 3 4 5 6 7 8

nums = { 2, 3, 1, 5, 4, 2, 1, 3, 6 }

L R V

Queries = [(1, 6, 2), (2, 5, 1) ....]

normal DAT:- diff = [0, 2, +1, 0, 0, 0, -1, -2, 0]

0 1 2 3 4 5 6 7 8

L = 1
R = 6
V = 2

diff[L] += V ;
diff[R+1] -= V ;

cumSum = [0, 2, 2, 2, 2, 2, 2, 0, 0]

nums = [2, 5, 3, 7, 6, 4, 3, 3, 6]

```
for (i = 0; i < n; i++) {
    if (i-1 >= 0) {
        diff[i] += diff[i-1];   → cum sum
    }
}
```

Expected = { 2, 5, 1, 7, 4, 4, 1, 3, 6 }

	0	1	2	3	4	5	6	7	8
nums =	2,	3,	1,	5,	4,	2,	1,	3,	6
		L					R		
diff =	0,	0,	0,	0,	0,	0,	0,	0,	0
		i							

L = 1

R = 6

V = +2

K = 2

```
diff[L] += V;
diff[R+1] -= V;
```

next

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
	0,	0,	0,	0,	0,	0,	0,	0,	0,	0,	0,	0,	0,	0,	0
		L							R						

val = 2

K = 5

diff[11] -= V

```
diff[i] += diff[i-1];   //cumsum
diff[i] += diff[i-K];
```

```
diff[L] += V ;
steps = (R-L)/K ;
next = l + (steps+1) * K ;

diff[next] -= V ;
```

Generic DAT

```
// cumsum
for ( i=0; i<n; i++) {
        if (i-K >=0) {
            diff[i] += diff[i-K] ;
        }
}
```

Queries = { (1, 6, 2, 2), (2, 8, 1, 2), (0, 5, 1, 3), (2, 6, 1, 3) . . . . . }

K=2 K=2 K=3 K=3

L R V K L R V K

nums = { 1, 2, 3, 2, 2, 1, 2, 4, 2, 5, 2, 1 }

0 1 2 3 4 5 6 7 8 9 10 11

$$\text{diff} = \{\overset{0}{0}, \overset{1}{+2}, \overset{2}{0}, \overset{3}{0}, \overset{4}{0}, \overset{5}{0}, \overset{6}{0}, \overset{7}{-2}, \overset{8}{0}, \overset{9}{0}, \overset{10}{0}, \overset{11}{0}\}$$

L K

steps = (R−L)/K = 2

next = L + (step+1) * K

= 1 + (3) * 2

= 1 + 6 = 7

L = 1

R = 6

v = +2

K = 2

diff[1] += 2

diff[7] -= 2

CumSum

K = 2 → { (1, 6, 2) , (2, 8, 1) }

L R v

K = 3 → { ( ) , ( ) }

K → Queries . DAT → nums

K' → Qu . DAT → nums

NOTE :-

If there are multiple Ks, group queries for each K and handle their queries at once.

$K \rightarrow \{Q1, Q2, Q3\} \rightarrow$ DAT

$K' \rightarrow \{Q4, Q5, Q6\} \rightarrow$ DAT $\rightarrow$ nums

⋮

What if Query asks for multiplication of val.

$Q = (L, R, V, K)$

$L - R$ = multiply each element by V

$(1, 4, \overset{V}{2})$

diff[L] *= V;
diff[next] /= V;

	0	1	2	3	4	5 (next)	...
diff =	1,	2	1,	1,	1,	1/2	...

// Cumulative product

```
for( i=0;  i<n ;  i++ ) {
        if(i-k >=0) {
            diff[i]  *= diff[i-k];
        }
}

// Apply diff arr  to nums
for( i=0;   i<n;  i++){
        nums[i]  *= diff[i];  %M ;
}
```

Leetcode → 3655

+

Squareroot decomposition